

26年5月第2套-题目文本-分享版(2)

SAT questions - standardized for website upload

Math Module 1

Question 3

Source PDF page: 2

A jar has 570 buttons, and 20% of these buttons are red. How many buttons in the jar are red? If the equations $9x + y = 20$ and $9x - y$ point (x, y) do the graphs intersect? 0 are graphed in the xy -plane, at what A. $(0, 20)$ B. $(0, 9)$ C. $(20, 0)$ D. $(20, 9)$ A square has side length of 47 inches. What is the area, in square inches, of the square? A. 2209 B. 188 A model estimates that the initial number of photons in an X-ray beam is 600 when the beam reaches the surface of a certain material. The model also estimates that when the beam passes through this material, the number of photons in the beam decreases by 50% for each 13 millimeters of the material the beam passes through. Which equation represents this model, where J is the estimated number of photons in the beam when the beam reaches a point x millimeters from the surface? A. $J = 60000$ B. $J = 60005$ C. $J = 600(2)^x$ D. $J = 600(3)^x$

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 1

Source PDF page: 3

The function f is defined by $f(x) = a$. What is the slope of the graph of $y = f(x)$ in the xy -plane?

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 8

Source PDF page: 4

The function g is defined by $g(x) = 16x + 25$. For what value of x does $g(x) = 297$? A group of 10 gardeners recorded data on the germination rates of their tomato crop for one growing season. (For additional SAT study materials, contact wechat: 13311268274.) The scatterplot shows the relationship between the number of tomato seeds planted, x , and the number of tomato seeds that germinated, y , for each of the gardeners. A line of best fit is also shown. Which of the following is the best interpretation of the slope of the line of best fit in this context? A. The number of tomato seeds planted is predicted to increase by 70 seeds every 100 days B. The number of tomato seeds planted is predicted to increase by 350 seeds every 100 days C. The number of tomato seeds that germinate is predicted to increase by 70 seeds for every additional 100 tomato seeds that are planted D. The number of tomato seeds that germinate is predicted to increase by 350 seeds for every additional 100 tomato seeds that are planted

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 1

Source PDF page: 5

The product of a positive number x and the number that is 8 more than x is 52. What is the value of x ?

- A. [choice text not detected]

- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 10

Source PDF page: 5

'Which expression is equivalent to $(x^2 + 22) + (x^2 + 3x^2)$?' A. $2x^2 + 22$ B. $2x^2 + 3x^2 + 22$ C. $2x^2 + 22 + 3x^2$ D. $2x^2 + 3x^2 + 22$

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 12

Source PDF page: 5

"The given equation relates the distinct positive variables x , y , and z . Which equation correctly represents z in terms of x and y ?" A. $z = x + y + 6$ B. $z = x + y - 6$ C. $z = x - y + 6$ D. $z = x - y - 6$

"The given function g is graphed in the xy -plane, where $y = g(x)$. What is the y -intercept of the graph?" A. $(0, 0)$ B. $(0, 3)$ C. $(3, 0)$ D. $(3, 4)$

"What is the sum of the solutions to the given equation?" A. -2 B. -1 C. 1 D. 2

In the xy -plane, circle M is the graph of the equation $(x - 2)^2 + (y - 6)^2 = 9$. Circle P has the same center as circle M but has a radius that is twice the radius of circle M . Which equation represents circle P ?

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 18

Source PDF page: 6

"The function f is defined by $f(x) = 2x + 3$. Which equation represents the graph of f ?"

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 16

Source PDF page: 7

A chef made 34 cups of sauce. The chef then filled x small jars and y large jars with all the sauce he made. (This SAT real test paper has been compiled by the HDF institution.) The equation $3x + 5y = 34$ represents this situation. Which is the best interpretation of $5y$ in this context?

A. The number of large jars the chef filled B. The number of small jars the chef filled C. The total number of cups of sauce in the large jars D. The total number of cups of sauce in the small jars

"The function j is defined by $j(2) = 8$ and the function k is defined by $k(x) = 4x$, where p is a positive constant. What is the value of p ?"

Line j is defined by $4x + 5y = 25$. Line k is parallel to line j in the xy -plane. An equation of line k is $16x - 20y = 15$, where T is a constant. If line k passes through the point $(0, 5)$, what is the value of T ?"

Note: Figure not drawn to scale. In triangles RST and XYZ shown, XV is parallel to TS and $\tan R = 2$. What is the value of $\sin X$ in triangle XYZ ?

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]

D. [choice text not detected]

Question 20

Source PDF page: 8

yy \$419 = 18 what is the value of $\frac{a}{b}$? D. -IT 'Triangle ABC is similar to triangle XYZ such that A, B, and C correspond to X, Y, and Z, respectively. (This SAT real test paper has been compiled by the HDF institution,) The length of each side of triangle ABC is m times the length of its corresponding side in triangle X $\neq$ Z, where m is an integer greater than 1. The measure of angle C is 58° , and $YZ = 71$. Which of the following must be true? B, The measure of angle Z is 58° . ©. BC = 7 In MASK BSE SRE are PERSATLOBAEANS of, wep OcAzsunean EENIED 0 sane2rsxnnuse, apes feenan © ROM He, SoMTSSRR SEED 0 coucsioemain, wmes, nase fsnemGnEm © RANE-2, cessaemeneN mary © sesirnsn GEEZID 0 pornianuss, exnemamere swenen © fnzORT ERR, MeN @ menusrasn SEEEIIOED 6 roraiianuso, sanaxanxe semen © RLANAMUNTANZSB, AAWERAO 5 HaSKTRMSTENRER (95) SEED 9 srxrsirsnuen (arene) Ssecren © SIRMCMRUREER (axxcKE nNE - eo sumawanans ERED 6 sstnisenwsso sKloRsaReeR SANEROR S sumesanic ton = © crmannaxe EEL 6 siisvzaannen eaemem © SAiuEAMENOON EXELIELD 0 stuiumanenoxsse Gacguen @S2EaAMENON - _ o sermratenene CEE 9 svinsncas Deena 92808 nencmanan EEELIELD 0 vesnzsxonex gsenen SSRI [=] JABS IAAY

A. [choice text not detected]

B. [choice text not detected]

C. [choice text not detected]

D. [choice text not detected]

Math Module 2

Question 5

Source PDF page: 12

"The function g is defined by $g(x) = 2x + 8$, where $g(a) = 18$ and $g(4) = b$. What is the value of $a + b$? saooremeasanam VE

A. [choice text not detected]

B. [choice text not detected]

C. [choice text not detected]

D. [choice text not detected]

Question 6

Source PDF page: 13

A right square prism has a height of 9 units. The volume of the prism is 324 cubic units. What is the length, in units, of an edge of the base? D. $3\sqrt{4}$ 15px = 60 In the given equation, p is a constant. The equation has no solution. What is the value of p ? 2 If $x^2 - 8$ represents all solutions to the inequality $ax - 12 < 36$, where a is a constant, what is the greatest possible value of a ? saooremeasanam VE In the given system of equations {Reach out to our assistant sat_800800 to receive 'more satellite learning resources.) p is a positive constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of xy ? D. 90 saooremeasanam VE

A. [choice text not detected]

B. [choice text not detected]

C. [choice text not detected]

D. [choice text not detected]

Question 10

Source PDF page: 15

ys $6x - 18$ ypc $4x - 11$ For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities? A B, xy xy | 7 5 | 8 [pas] 6 | i 3 | -6 28 c D. FP] sly 10 (2. 0 16 -8 | 28 6 ease

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 11

Source PDF page: 16

'The function f is defined by $f(x) = 41(0.26)^x$. For any positive integer m , the value of $F()$ is $pM\%$ less than the value of $(7m \sim 1)$. What is the value of p ? D.26

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 12

Source PDF page: 16

A community organization surveyed a random sample of residents of the community to estimate the percentage of residents who visit their local library at least once per 'month, From the survey, the organization estimates that 1296 of residents of the community visit their local library at least once per month, with an associated margin 'of error of 3.7%, The organization repeated the survey with a random sample of residents that was double the original sample size. Assuming the margins of error are calculated in the same way far both studies, which of the following is true about the 'margin of error associated with the estimate from the larger sample? A. The margin of error is less than 3.7%, B, The margin of error is 7.496. . The margin of error is greater than 12%, D. The margin of error is 24%. gene saorassanae TE)

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 13

Source PDF page: 17

'The line shown can be represented by the equation $ax + by = 48$, where a and b are constants, Which of the following is true about a and b ? 4
 $a < 0$ and $b < 0$ B. $a < 0$ and $b > 0$ © $a > 0$ and $b < 0$ D. $a > 0$ and $b > 0$ ~3x2 + bx 48=0 In the given equation, 6 is a positive integer. The equation has no real solution. What is the largest possible value of & ? nuosmeasesmnm UP

- A. [choice text not detected]
- B. [choice text not detected]
- C. [choice text not detected]
- D. [choice text not detected]

Question 15

Source PDF page: 18

Each 3.0 -ounce serving of cheddar cheese and each 1.2 -ounce serving of tuna provides about 1 microgram of vitamin B12. (This SAT real test paper has been compiled by the HDF institution.) If a total of 1.8 micrograms of vitamin B12 are consumed from eating x ounces of cheese and y ounces of tuna, which equation best represents this situation? A, $0x + 12y = 18$ B. $12x + 3.0y = 18$ € $0.83x + 0.33y = 18$ D. $0.38x + 0.83y = 1.8$

- A. [choice text not detected]
- B. [choice text not detected]

C. [choice text not detected]

D. [choice text not detected]

Question 16

Source PDF page: 18

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A. [choice text not detected]

B. [choice text not detected]

C. [choice text not detected]

D. [choice text not detected]

Question 17

Source PDF page: 19

A zoologist observed the nests of wood ducks in an area. Each year, the zoologist recorded the number of eggs in each of the first 40 nests that were observed and created a frequency table for the data set. Which of the following frequency tables represents the data set with the smallest standard deviation? A. B. C. D.

A. [choice text not detected]

B. [choice text not detected]

C. [choice text not detected]

D. [choice text not detected]

Question 19

Source PDF page: 20

'The area of a rectangular region is increasing at a rate of 140 square feet per hour. 'Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet) B. 0.71 C. 1.65 D. 13.01

A. [choice text not detected]

B. [choice text not detected]

C. [choice text not detected]

D. [choice text not detected]

Question 20

Source PDF page: 20

'A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 2400% of the number of prey at the start of the week, and the number of predators had increased by 450% of the number of predators at the start of the week. The number of predators at the end of the week was 7% less than the number of prey at the end of the week. What is the value of p? gene saaorassanae TE) Te - 55x-72=0 'What is the positive solution to the given equation? 'The function f is a quadratic function. In the xy-plane, the graph of y = f(x) has a vertex at (1,9) and passes through the points (2,37) and (-1,121). What is the value of f(-2) - f(0)? c. 24 D. 261 saoremeasanam VE

A. [choice text not detected]

B. [choice text not detected]

C. [choice text not detected]

D. [choice text not detected]